Yiqi (Michael) Yao

Machine Learning Engineer & AI Researcher

Status: B.S. in Joint CS & Mathematics, Harvey Mudd College, GPA 3.90 (Expected May 2028) — Claremont, CA
Focus: LLM Post-Training (RL), Tool-Use Agents, AI for Science, Mechanistic Interpretability — michaelyao.site
Stack: Python, C++, SQL, PyTorch, verl, GRPO/PPO, FSDP, vLLM, MCP, TabPFN, Docker, Linux/HPC — miyao@hmc.edu
Links: github.com/OhhMoo — 909-407-5410

Industry Experience

Machine Learning Engineering Intern - Alkera AI (YC S26), San Francisco, CA — May 2026 - Present

Execution-Verified Reinforcement Learning

- Implemented **multi-turn GRPO** in `verl` for SQL pipeline synthesis and debugging with iterative tool execution
- Scaled **Qwen3.5 9B to 27B** (8×A100 to 8×H100); raised mean execution-graded score **67%** (0.18 to 0.30)
- Bounded **execution-verifier rewards** with baseline and gold anchors to reward correct SQL outcomes over surface matches

Sandboxed Agent Infrastructure & Evaluation

- Built a sandboxed **63-tool MCP gateway** emulating **19 vendor platforms**, integrating DuckDB, dbt, BI, Jira, and Slack
- Designed **32 data-incident tasks** with deterministic execution graders and tiered rewards; evaluated **483 agent rollouts**
- Patched **sandbox-escape and root-credential exploits** uncovered in agent rollouts to prevent execution-based reward hacking

R&D Data & Quality Systems Intern - Fresenius Kabi, Wuxi, China — Aug 2026 - Sep 2026

- Built a **Python and JavaScript ETL** pipeline to automate quality-compliance audits across **665 documents**, reconciling 222 SOP records against master databases and automatically repairing **189 broken cross-references**

Research Experience

Researcher (Advisor: Bilin Zhuang) - Harvey Mudd College — Oct 2025 - Present

Unsupervised Machine Learning on Physical System

- Developed **Gaussian mixture classification** in $(q, \mathrm{LSI}, s_k, \zeta)$ space with **likelihood-ratio abstention**
- Analyzed local structural populations across **TIP4P-2005, TIP5P, and SWM4-NDP** molecular dynamics trajectories
- Computed **state-resolved structure factors**, resolving tetrahedral and disordered signatures that **converge upon heating** [2]

Mean-Field Theory of Deep Neural Networks

- Derived **mean-field variance** and **inter-input correlation recursions** for signal propagation through deep tanh networks
- Characterized the **order-to-chaos boundary** ($\chi_1 = 1$); validated theoretical recursions against **width-1,000 simulations**

ML Researcher (Advisor: Miquel Duran-Frigola) - Ersilia.io — Apr 2026 - Present

- Benchmarked molecular representations across **77 few-shot tasks**; broad RDKit features improved mean ROC-AUC by **3.5–5.3 points** [1]
- Trained **238 source-assay predictors** and an assay-selection model on **24,675** draw-averaged **TabPFN** evaluations
- Applied **equal-width SVD** and **paired-removal controls**; learned weighting added 0.6–0.9 ROC-AUC points at higher inference cost
- Audited 9 ADME/Tox benchmarks for **source-training leakage**; identified near-total query/source overlap in **3 endpoints**
- Released the open-source **ChemICL** prediction client with provenance screening and optional **GLiNER2** model-card extraction [1]

Projects

NullStep - SAE Feature Drift in Fine-Tuned LLMs - with Cole Blodin (AWS) | PyTorch, SAELens, PPO — Jan 2026 - Present

- Introduced a **"null-step" control** with fixed model weights to isolate **dictionary retraining noise** in SAE feature-drift metrics [3]
- Evaluated **112 TopK and JumpReLU SAEs** across **SFT and PPO** checkpoints to quantify feature-drift confounding
- Separated fine-tuning-induced **representation drift** from **SAE optimization instability** through comparisons against null-step controls

LangAlpha - Multi-Agent Financial Research Platform - Python, TypeScript, FastAPI, SSE — Nov 2025 - Jan 2026

- Implemented real-time **Server-Sent Events (SSE)** streams to expose agent reasoning in a multi-agent financial research platform
- Built **per-turn source citation tracking** to link agent-generated research to supporting evidence [Repo]

Publications & Honors

- **[1] Y. Yao**, M. Duran-Frigola. *Feature Selection for Tabular Foundation Models: Exploring Feature Utility in Low-Data Molecular Prediction*. arXiv:2609.30789, 2026.
- **[2] Y. Yao**, D. Sanghi, A. D. Chokshi, et al. *Is Water a Mixture: Unsupervised Likelihood-Ratio Classification Reveals Two Water States That Merge Upon Increasing Temperature*. Under review, *PNAS*, 2026. [Code]
- **[3] Y. Yao**, C. Blodin, et al. *A Null-Step Control for Sparse-Autoencoder Feature Drift Across Fine-Tuning Checkpoints*. Manuscript, 2026. [Weights]
- **Olympiad Honors**: UKChO Gold (2023, 2024), Silver (2022); CCO Round 2 Gold (2024), Round 1 Gold (2022, 2023); PhysicsBowl Gold (Top 10%)